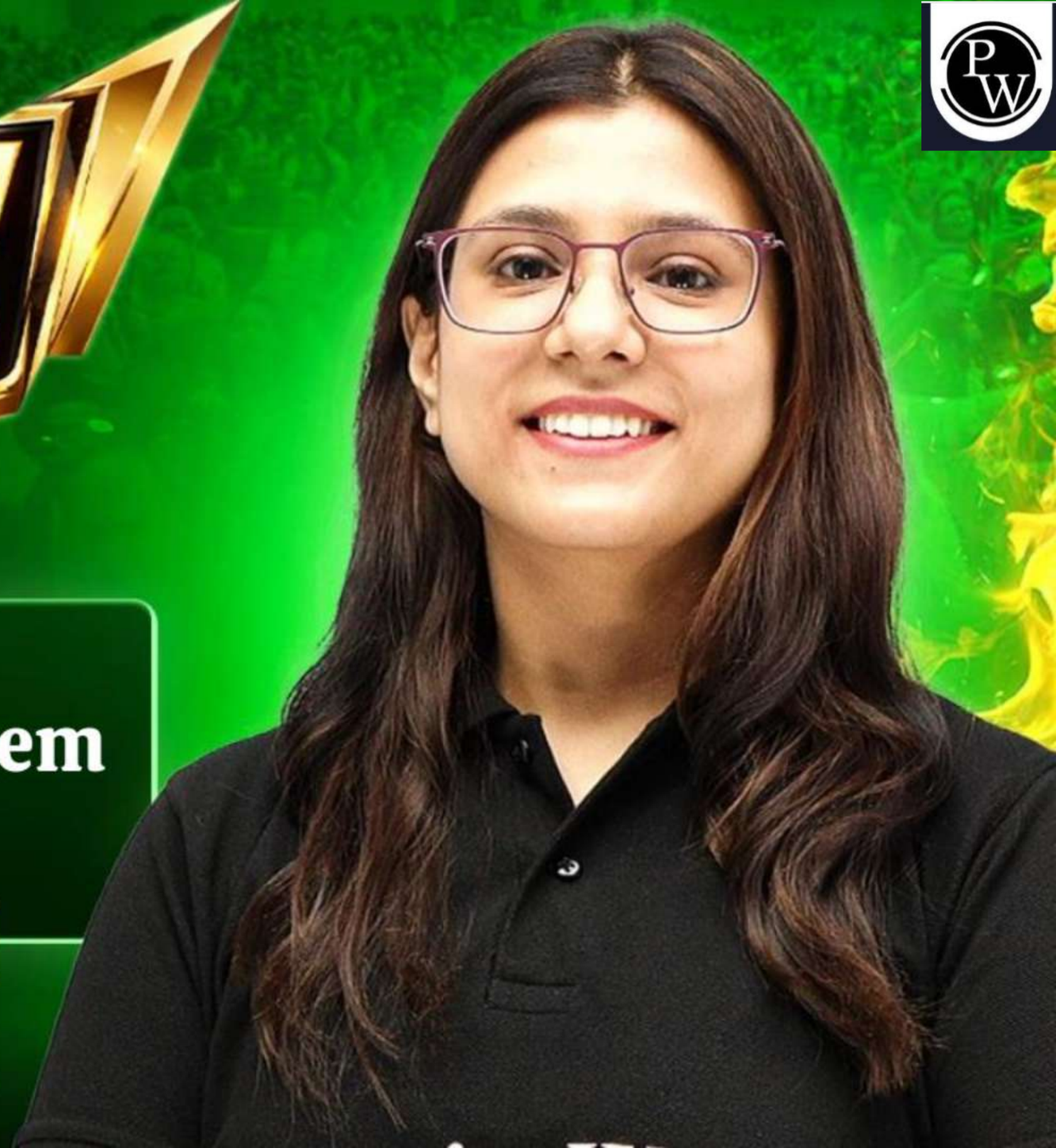
VICTORY 2027

The Circulatory System

LECTURE-5 BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. Blood Vessels
2. Structure of Human Heart
3.

RATTON KO KARO BYE BYE !!

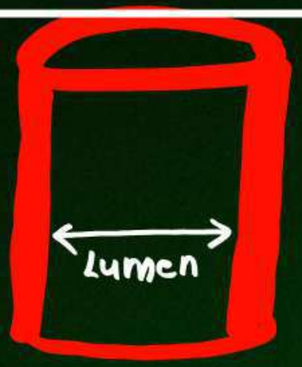
EASY CONCEPTS SE KARO HELLO HI !!

With – Uppu Ma'am



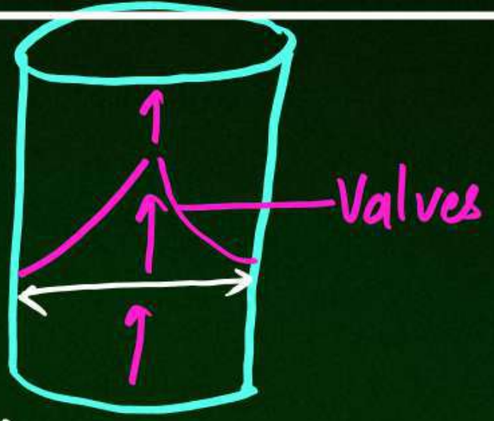
BLOOD VESSELS

3 type → Arteries, Veins, Capillaries



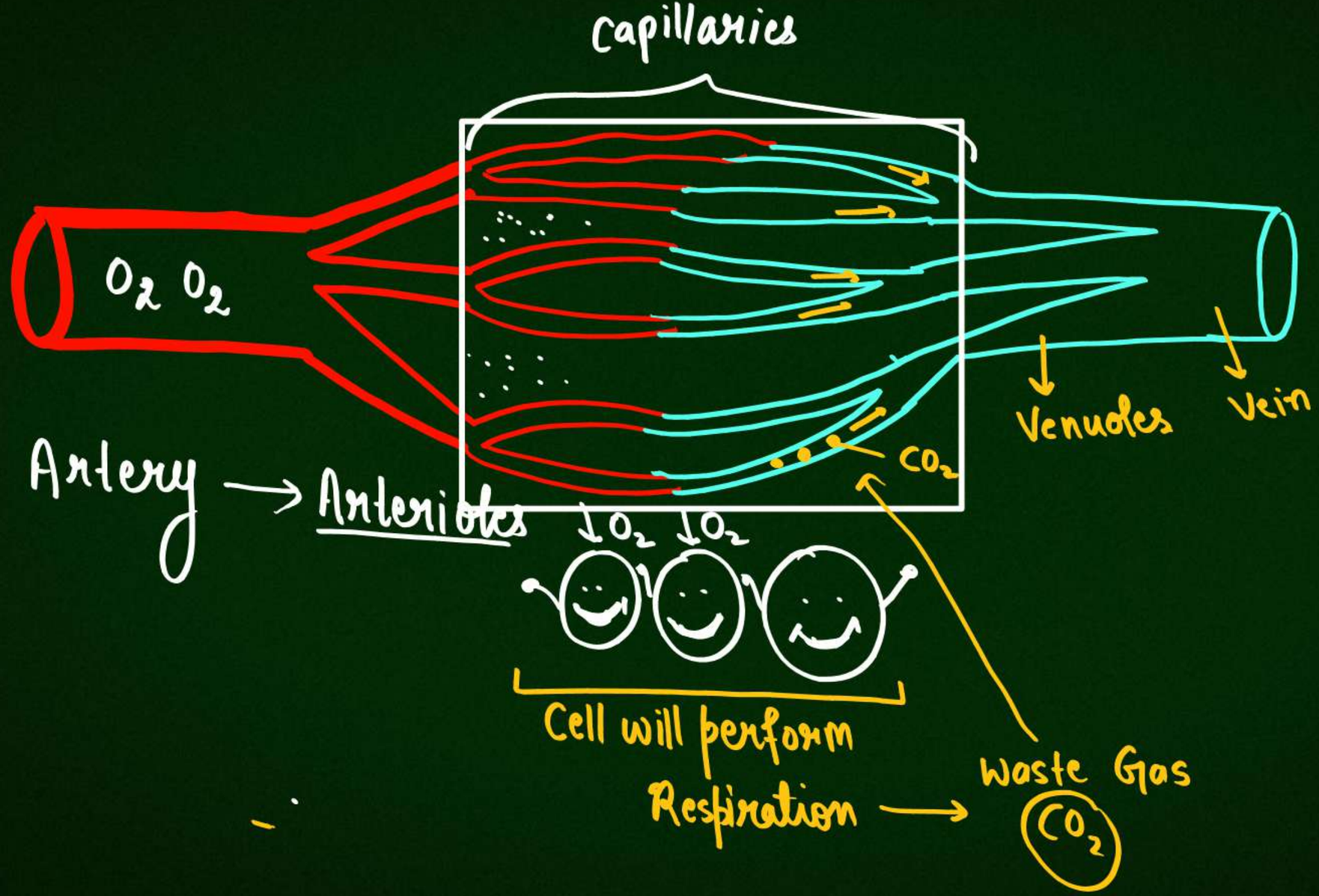
Artery

- Thick Wall
- Elastic
- Narrow Lumen
- Carry blood away from the ♡
- Blood with high pressure.
- Valves are absent
- Always carry Oxygenated Blood (O_2-Hb)
- Exception
  PULMONARY ARTERY [carries deoxy. Blood]
  Lungs
- Deeply seated



Vein

- Thin walls
- Non Elastic
- Wide lumen
- Brings blood back to the ♡
- Blood flow with low pressure.
- Valves are present, to prevent the backflow of blood
- Always carry deoxygenated blood (CO_2-Hb)
- Pulmonary Vein [carries O_2Hb]
- Superficially placed

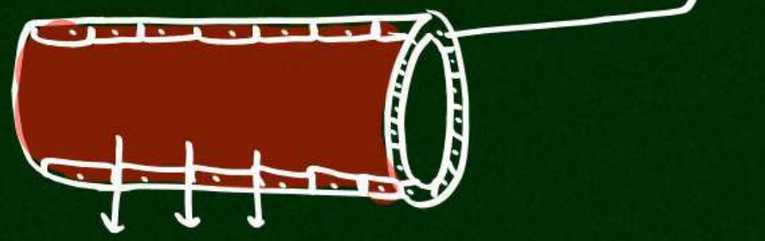

Capillaries
O_2 O_2
Artery → Arterioles
Venuoles
Vein
CO_2
$\downarrow O_2$ $\downarrow O_2$
Cell will perform Respiration →
Waste Gas
CO_2

Capillaries

→ Capillaries help in exchange of gases.

→ The size of capillaries is very small.

→ Lined by single wall of **Squamous epithelial tissue.**



ENDOTHELIUM

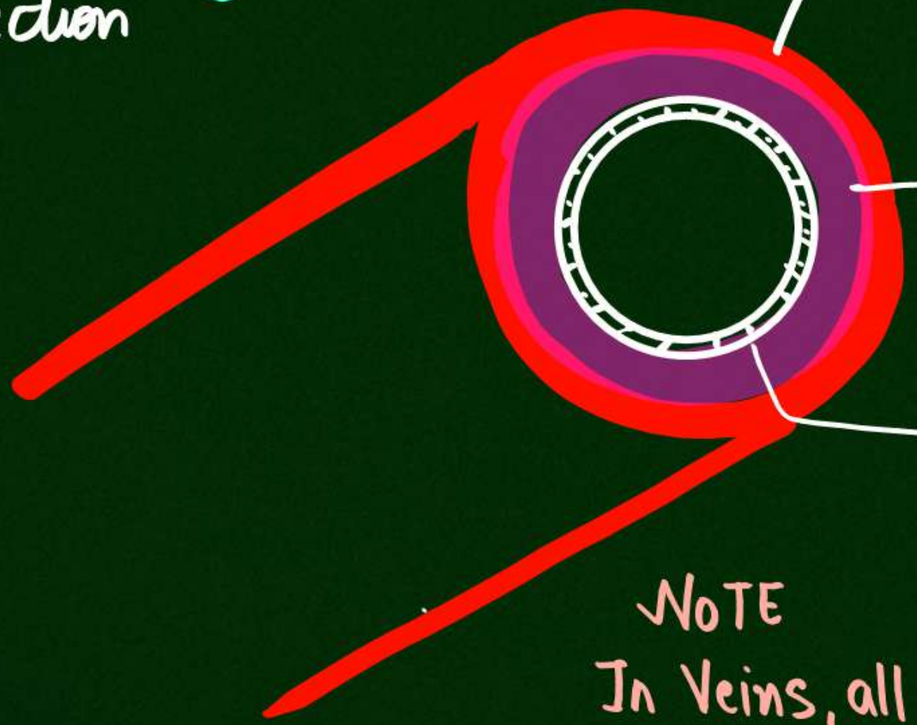
- It is the wall of capillary

2023, 2024, 2025

Diagram based Question

T.S of Artery

Transverse Section



Tunica Externa [fibrous epithelial connective tissue]

Tunica Media [smooth muscles (involuntary)]

Tunica Interna [squamous Epithelial tissue]

NOTE

In Veins, all 3 layers are present but Tunica media is thin.

Arteries can constrict (vasoconstriction) and dilate (vasodilation) to increase or decrease the blood pressure.

Q. The blood clotting is caused by which enzyme.

A) ThromboKinase

B) Stuart factor

C) Prothrombin

D) All of the above ✓

QUESTIONS

The blood vessels carry blood from the heart to the various parts of the body are called _____.

(A) Arteries

(B) Veins

(C) Septum

(D) More than one of the above

QUESTIONS

Why are veins shown as blue in diagrams, though actual blood is dark red?

A. To indicate oxygen-rich blood

B. To indicate warmer blood

C. For contrast against arteries

D. Because the blood is truly blue inside veins

Thank You